Linear System — Homework 5

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Exercise 1

Suppose A is a 3×3 matrix and

$$\det(sI - A) = s^3 + 2s^2 + s + 1.$$

(a) Express A^3 in terms of I, A, and A^2.

Solution

$$\begin{aligned} p(s) &= \det(sI - A) = s^3 + 2s^2 + s + 1 \\ p(A) &= A^3 + 2A^2 + A + I = 0 \\ \Rightarrow A^3 &= -2A^2 - A - I \end{aligned}$$

(b) Express A^5 in terms of I, A, and A^2.

Solution

$$A^5 = A^2 A^3 = -2A^4 - A^3 - A^2$$

$$\begin{aligned} A^4 = A^3 A &= -2A^3 - A^2 - A \\ &= -2(-2A^2 - A - I) - A^2 - A \\ &= 4A^2 + 2A + 2I - A^2 - A \\ &= 3A^2 + A + 2I \end{aligned}$$

$$\begin{aligned} A^5 &= -2(3A^2 + A + 2I) - (-2A^2 - A - I) - A^2 \\ &= -6A^2 - 2A - 4I + 2A^2 + A + I - A^2 \\ &= -5A^2 - A - 3I. \end{aligned}$$

(c) Express A^{-1} in terms of I, A, and A^2.

Solution

Since $p(0) = \det(-A) = 1$ and A is 3×3, $\det A = -1 \neq 0$. Thus A is invertible.

$$\begin{aligned} A^{-1}p(A) &= A^2 + 2A + I + A^{-1} = 0 \\ \Rightarrow A^{-1} &= -A^2 - 2A - I \end{aligned}$$

Exercise 2

Without doing any matrix multiplications, compute A^4 for

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 564 & 0 & 0 & 0 \end{bmatrix}.$$

Justify your answer.

Solution

$$\begin{aligned} p(s) = \det(sI - A) &= \begin{vmatrix} s & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & -1 \\ -564 & 0 & 0 & s \end{vmatrix} \\ &= s \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 0 & s \end{vmatrix} - (-564) \begin{vmatrix} -1 & 0 & 0 \\ s & -1 & 0 \\ 0 & s & -1 \end{vmatrix} \\ &= s^4 - 564 \end{aligned}$$

$$\begin{aligned} \Rightarrow p(A) &= A^4 - 564I = 0 \quad \text{(Cayley–Hamilton theorem)} \\ \therefore A^4 &= 564I \end{aligned}$$

Exercise 3

Consider the differential equation

$$\dot{x} = Ax,$$

where A is a square matrix. Show that if A has $j2\pi$ as an eigenvalue, then there is a nonzero initial state x_0 such that the solution x satisfies

$$x(1) = x(0) = x_0.$$

Solution

Since $j2\pi$ is an eigenvalue of A, choose a nonzero eigenvector v such that $Av = j2\pi v$. Set $x_0 = v$ and let

$$x(t) = e^{j2\pi t}v.$$

Then $x(0) = v = x_0$ and

$$\dot{x}(t) = j2\pi e^{j2\pi t}v = e^{j2\pi t}Av = Ax(t).$$

$$\Rightarrow x(1) = e^{j2\pi}x_0 = x_0 = x(0)$$

If A is real and a real initial state is required, take the real or imaginary part of this solution whose initial value is nonzero. Both parts satisfy $\dot{x} = Ax$ and return to their initial values at $t = 1$.

Exercise 4

Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

by:

(a) use Jordan form using eigenvalues and eigenvectors of A;

Solution

$$\det(sI - A) = \begin{vmatrix} s & -1 \\ 1 & s \end{vmatrix} = s^2 + 1 = 0, \quad \therefore s = \pm j$$

for $s = j$

$$\begin{bmatrix} j & -1 \\ 1 & j \end{bmatrix} v = \begin{bmatrix} j & -1 \\ 1 & j \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} j \\ -1 \end{bmatrix}$$

for $s = -j$

$$\begin{bmatrix} -j & -1 \\ 1 & -j \end{bmatrix} v = \begin{bmatrix} -j & -1 \\ 1 & -j \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} j \\ 1 \end{bmatrix}$$

Therefore

$$T = \begin{bmatrix} j & j \\ -1 & 1 \end{bmatrix}, \quad J = \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix}, \quad T^{-1} = \frac{1}{2j} \begin{bmatrix} 1 & -j \\ 1 & j \end{bmatrix}$$

By $A = TJT^{-1}$

$$e^{At} = T \begin{bmatrix} e^{jt} & 0 \\ 0 & e^{-jt} \end{bmatrix} T^{-1}$$

Hence, at $t = \ln 2$

$$e^{A\ln 2} = T\begin{bmatrix} e^{j\ln 2} & 0 \\ 0 & e^{-j\ln 2} \end{bmatrix} T^{-1} = \frac{1}{2j}\begin{bmatrix} j & j \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 2^j & 0 \\ 0 & 2^{-j} \end{bmatrix}\begin{bmatrix} 1 & -j \\ 1 & j \end{bmatrix}$$
$$= \frac{1}{2}\begin{bmatrix} 2^{-j}+2^j & -j(2^j-2^{-j}) \\ j(2^j-2^{-j}) & 2^{-j}+2^j \end{bmatrix} = \begin{bmatrix} \cos(\ln 2) & \sin(\ln 2) \\ -\sin(\ln 2) & \cos(\ln 2) \end{bmatrix}$$

(b) using the Laplace transform.

Solution

By

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1} \quad \Leftrightarrow \quad e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

$$(sI - A)^{-1} = \begin{bmatrix} s & -1 \\ 1 & s \end{bmatrix}^{-1}$$
$$= \begin{bmatrix} \frac{s}{s^2+1} & \frac{1}{s^2+1} \\ -\frac{1}{s^2+1} & \frac{s}{s^2+1} \end{bmatrix}$$
$$\Rightarrow \mathcal{L}^{-1}\{(sI - A)^{-1}\} = e^{At} = \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{s}{s^2+1} & \frac{1}{s^2+1} \\ -\frac{1}{s^2+1} & \frac{s}{s^2+1} \end{bmatrix}\right\} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

Hence, at $t = \ln 2$

$$e^{At} = \begin{bmatrix} \cos(\ln 2) & \sin(\ln 2) \\ -\sin(\ln 2) & \cos(\ln 2) \end{bmatrix}$$

Exercise 5

Compute e^{At} for

$$A = \begin{bmatrix} 3 & -5 \\ -5 & 3 \end{bmatrix}$$

by:

(a) use Jordan form using eigenvalues and eigenvectors of A;

Solution

$$\det(sI - A) = \det\begin{bmatrix} s-3 & 5 \\ 5 & s-3 \end{bmatrix} = s^2 - 6s - 16 = (s-8)(s+2)$$

therefore, eigenvalues $s = 8$ and $s = -2$

for $s = 8$

$$(sI - A)v = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

for $s = -2$

$$(sI - A)v = \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad J = \begin{bmatrix} 8 & 0 \\ 0 & -2 \end{bmatrix}, \quad T^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Finally

$$\begin{aligned} e^{At} &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} e^{8t} & 0 \\ 0 & e^{-2t} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} e^{8t} & e^{-2t} \\ -e^{8t} & e^{-2t} \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} e^{8t} + e^{-2t} & -e^{8t} + e^{-2t} \\ -e^{8t} + e^{-2t} & e^{8t} + e^{-2t} \end{bmatrix} \\ &= \frac{e^{3t}}{2}\begin{bmatrix} e^{5t} + e^{-5t} & -(e^{5t} - e^{-5t}) \\ -(e^{5t} - e^{-5t}) & e^{5t} + e^{-5t} \end{bmatrix} \\ &= e^{3t}\begin{bmatrix} \cosh(5t) & -\sinh(5t) \\ -\sinh(5t) & \cosh(5t) \end{bmatrix} \end{aligned}$$

(b) using the Laplace transform.

Solution

By

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1} \Rightarrow e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

$$(sI - A)^{-1} = \frac{1}{(s-8)(s+2)}\begin{bmatrix} s-3 & -5 \\ -5 & s-3 \end{bmatrix}.$$

For the $(1,1)$ entry:

$$\frac{s-3}{(s-8)(s+2)} = \frac{1}{2}\frac{1}{s-8} + \frac{1}{2}\frac{1}{s+2}.$$

For the $(1,2)$ entry:

$$\frac{-5}{(s-8)(s+2)} = -\frac{1}{2}\frac{1}{s-8} + \frac{1}{2}\frac{1}{s+2}.$$

For the $(2,1)$ entry:

$$\frac{-5}{(s-8)(s+2)} = -\frac{1}{2}\frac{1}{s-8} + \frac{1}{2}\frac{1}{s+2}.$$

For the $(2,2)$ entry:

$$\frac{s-3}{(s-8)(s+2)} = \frac{1}{2}\frac{1}{s-8} + \frac{1}{2}\frac{1}{s+2}.$$

$$\begin{aligned}\Rightarrow \mathcal{L}^{-1}\{(sI-A)^{-1}\} &= \frac{1}{2}\begin{bmatrix} e^{8t}+e^{-2t} & -e^{8t}+e^{-2t} \\ -e^{8t}+e^{-2t} & e^{8t}+e^{-2t}\end{bmatrix} \\ &= e^{3t}\begin{bmatrix} \cosh(5t) & -\sinh(5t) \\ -\sinh(5t) & \cosh(5t)\end{bmatrix}\end{aligned}$$

Exercise 6

Compute e^{At} at $t = \ln(2)$ for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0\end{bmatrix}.$$

Solution

$$\det(sI-A) = \begin{vmatrix} s & -1 \\ 1 & s\end{vmatrix} = s^2+1 = 0, \quad \therefore s = \pm j$$

for $s = j$

$$\begin{bmatrix} j & -1 \\ 1 & j\end{bmatrix} v = \begin{bmatrix} j & -1 \\ 1 & j\end{bmatrix}\begin{bmatrix} v_1 \\ v_2\end{bmatrix} = \begin{bmatrix} 0 \\ 0\end{bmatrix} \quad \Rightarrow \begin{bmatrix} v_1 \\ v_2\end{bmatrix} = \begin{bmatrix} j \\ -1\end{bmatrix}$$

for $s = -j$

$$\begin{bmatrix} -j & -1 \\ 1 & -j\end{bmatrix} v = \begin{bmatrix} -j & -1 \\ 1 & -j\end{bmatrix}\begin{bmatrix} v_1 \\ v_2\end{bmatrix} = \begin{bmatrix} 0 \\ 0\end{bmatrix} \quad \begin{bmatrix} v_1 \\ v_2\end{bmatrix} = \begin{bmatrix} j \\ 1\end{bmatrix}$$

Therefore

$$T = \begin{bmatrix} j & j \\ -1 & 1\end{bmatrix}, \quad J = \begin{bmatrix} j & 0 \\ 0 & -j\end{bmatrix}, \quad T^{-1} = \frac{1}{2j}\begin{bmatrix} 1 & -j \\ 1 & j\end{bmatrix}$$

By $A = TJT^{-1}$

$$e^{At} = T\begin{bmatrix} e^{jt} & 0 \\ 0 & e^{-jt}\end{bmatrix}T^{-1}$$

Hence, at $t = \ln 2$

$$e^{A\ln 2} = T\begin{bmatrix} e^{j\ln 2} & 0 \\ 0 & e^{-j\ln 2} \end{bmatrix} T^{-1} = \frac{1}{2j}\begin{bmatrix} j & j \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 2^j & 0 \\ 0 & 2^{-j} \end{bmatrix}\begin{bmatrix} 1 & -j \\ 1 & j \end{bmatrix}$$
$$= \frac{1}{2}\begin{bmatrix} 2^{-j}+2^j & -j(2^j-2^{-j}) \\ j(2^j-2^{-j}) & 2^{-j}+2^j \end{bmatrix} = \begin{bmatrix} \cos(\ln 2) & \sin(\ln 2) \\ -\sin(\ln 2) & \cos(\ln 2) \end{bmatrix}$$

Exercise 7

Obtain (by hand) the state response and output response $x(\cdot)$ of the following system due to a unit impulse input and the zero initial conditions. Determine whether the state response contains all the system modes.

$$\begin{aligned} \dot{x}_1 &= -5x_1 + 2x_2 + u, \\ \dot{x}_2 &= -12x_1 + 5x_2 + u, \\ y &= 2x_1 + 3x_2. \end{aligned}$$

Solution

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} -5 & 2 \\ -12 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u = Ax + Bu \\ y &= \begin{bmatrix} 2 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = Cx + Du \end{cases}$$
$$\dot{x} - Ax(t) = Bu(t)$$

Multiply both sides on the left by e^{-At}. Since A commutes with e^{-At},

$$e^{-At}\dot{x}(t) - e^{-At}Ax(t) = e^{-At}Bu(t) \Rightarrow \frac{d}{dt}(e^{-At}x(t)) = e^{-At}Bu(t)$$

Integrate from 0^- to t:

$$e^{-At}x(t) - x(0^-) = \int_{0^-}^{t} e^{-A\tau}Bu(\tau)d\tau$$

Therefore

$$x(t) = e^{At}x(0^-) + e^{At}\int_{0^-}^{t} e^{-A\tau}Bu(\tau)d\tau$$

For zero initial conditions, $x(0^-) = 0$, and a unit impulse input, $u(t) = \delta(t)$,

$$\begin{aligned} x(t) &= e^{At}\int_{0^-}^{t} e^{-A\tau}B\delta(\tau)d\tau \\ &= e^{At}B, \quad t>0 \end{aligned}$$

and

$$y = Cx \Rightarrow y = Ce^{At}B, \quad t>0.$$

Solving e^{At}

$$A = \begin{bmatrix} -5 & 2 \\ -12 & 5 \end{bmatrix}$$

$$\begin{aligned} \det(sI-A) &= \begin{vmatrix} s+5 & -2 \\ 12 & s-5 \end{vmatrix} \\ &= s^2 - 25 + 24 = s^2 - 1 = (s+1)(s-1) = 0 \end{aligned}$$

Hence, A has two eigenvalues, 1 and -1.

for $s = 1$

$$(sI-A)v = \begin{bmatrix} 6 & -2 \\ 12 & -4 \end{bmatrix} v = 0 \Rightarrow v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

for $s = -1$

$$(sI-A)v = \begin{bmatrix} 4 & -2 \\ 12 & -6 \end{bmatrix} v = 0 \Rightarrow v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore

$$T = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}, \quad J = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad T^{-1} = \frac{1}{-1}\begin{bmatrix} 2 & -1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}$$

$$\begin{aligned} e^{At} = Te^{Jt}T^{-1} &= \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{bmatrix}\begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \\ &= \begin{bmatrix} e^{t} & e^{-t} \\ 3e^{t} & 2e^{-t} \end{bmatrix}\begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -2e^{t}+3e^{-t} & e^{t}-e^{-t} \\ -6e^{t}+6e^{-t} & 3e^{t}-2e^{-t} \end{bmatrix} \end{aligned}$$

Finally,

$$\begin{aligned} x(t) &= e^{At}B \\ &= \begin{bmatrix} -2e^t + 3e^{-t} & e^t - e^{-t} \\ -6e^t + 6e^{-t} & 3e^t - 2e^{-t} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} (-2e^t + 3e^{-t}) + (e^t - e^{-t}) \\ (-6e^t + 6e^{-t}) + (3e^t - 2e^{-t}) \end{bmatrix} \\ &= \begin{bmatrix} -e^t + 2e^{-t} \\ -3e^t + 4e^{-t} \end{bmatrix}, \qquad t > 0. \end{aligned}$$

and

$$\begin{aligned} y(t) &= Ce^{At}B \\ &= \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -e^t + 2e^{-t} \\ -3e^t + 4e^{-t} \end{bmatrix} \\ &= 2(-e^t + 2e^{-t}) + 3(-3e^t + 4e^{-t}) \\ &= -11e^t + 16e^{-t}, \qquad t > 0. \end{aligned}$$

The state response can be written as

$$x(t) = -\begin{bmatrix} 1 \\ 3 \end{bmatrix} e^t + 2\begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-t}.$$

Both modes e^t and e^{-t}, corresponding to the eigenvalues 1 and -1, have nonzero coefficients. Therefore, the state response contains all the system modes.